BlueBoat Controller Design

PID, LQR, and pole placement on the nonlinear simulator

J.C. Vaught October 11, 2026

This study implements three controllers for the most complete BlueBoat simulator and compares them with fixed tuning. All controllers track a straight east-going line at a surge speed of 1 m/s. Wind, gusts, prescribed wave forcing, nonlinear hull motion, installed M200 propulsion, motor dynamics, command timing, sensor errors, and added inertia remain enabled in the full-model tests.

- The task is moving line tracking.
  - Recover from a 2 m northward displacement, a 0.12 rad heading error, and an initial surge speed of 0.8 m/s.
    - North position should approach zero while surge speed approaches 1 m/s. East position is free; there is no timed arrival or along-track position target.
- The comparison uses the same feedback and actuator boundaries.
  - Delivered global navigation satellite system (GNSS) and inertial measurement unit (IMU) records feed a common timestamp-aware state estimate.
    - Camera observations remain simulated but are unused by these navigation controllers. Perfect-state feedback is enabled only in the explicitly named oracle ablation.
- Tuning is frozen before the comparison sweep.
  - 171 runs cover 19 cases and 3 paired seeds per method.
    - The results describe this simulator and operating envelope. The simulator report documents which physical coefficients require boat measurements.

> The controller named LQR includes integral augmentation and is therefore a linear-quadratic-integral regulator. Pole placement uses the same augmented states. Their integral terms let sustained disturbances produce a corrective bias rather than a permanent line error.

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1 Control objective and shared architecture

The two fixed thrusters generate surge force and yaw moment. They provide no direct sway force, so the vessel corrects cross-track error by changing heading while moving forward. This makes forward line tracking a suitable first comparison; zero-speed station keeping needs a different design.

- The reference consists of surge speed $U = 1$ m/s and line north coordinate $N_d = 0$ m.
  - Define $e_y = N - N_d$, $\delta u = u - U$, and counterclockwise heading ψ from east.
    - A northward wind can require a southward crab heading even when e_y and δu are near zero. Heading error against east is therefore not a suitable primary tracking score.
- The control outputs are two pulse-width modulation (PWM) commands.
  - A desired total thrust S_c and yaw moment M_c become individual force requests.
    - Port and starboard thrust are clipped independently to the installed M200 bounds. Both methods of state feedback and PID use the same allocation.

The allocation follows from the motor separation $b = 0.75$ m. Positive yaw requires greater starboard thrust.

$$F_{p,c} = \frac{S_c}{2} - \frac{M_c}{b}, \quad F_{s,c} = \frac{S_c}{2} + \frac{M_c}{b}. \tag{1}$$

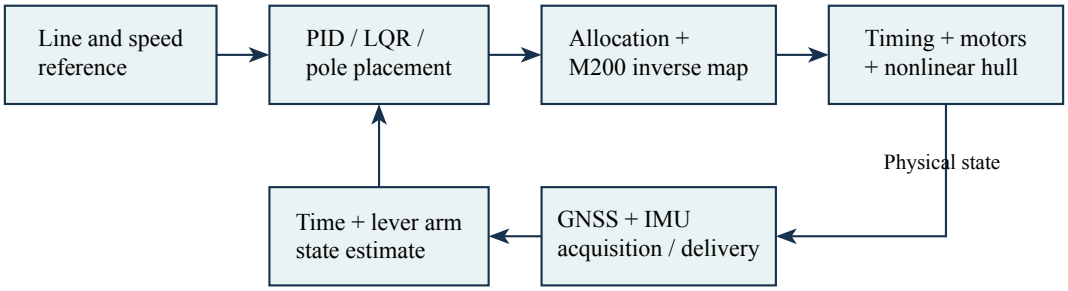


Figure 1: Shared feedback and actuation chain.

- The hardware boundary remains the one documented in the simulator report.
  - AGX Orin execution uses a 50 Hz decision tick and the configured 5 ms computation, 10 ms transport, and 20 ms actuator dead time.
    - These are simulation timing assumptions. The controller code does not benchmark the selected computer.
  - The navigation feedback uses GNSS position, east/north velocity, and dual-antenna heading, plus IMU yaw rate.
    - A camera is not needed for this line reference. Camera occlusion therefore has no control effect in this experiment.

1.1 Causal state estimate

The controllers need body-frame velocities and a current hull position. GNSS measures a displaced antenna and arrives late, so the shared estimator first corrects its lever arm and then extrapolates the corrected measurement to the decision time. It retains the most recent valid record through random dropouts.

The antenna offset is $\ell = -0.5$ m along the forward axis. At measurement heading ψ_g and yaw rate estimate $\hat{r}$, the hull origin and world velocity satisfy the following expressions.

$$\widehat{E}_g = E_a - \ell \cos\psi_g, \quad \widehat{N}_g = N_a - \ell \sin\psi_g. \tag{2}$$

$$\widehat{V}_E = V_{E,a} + \ell \hat{r} \sin\psi_g. \tag{3}$$

$$\widehat{V}_N = V_{N,a} - \ell \hat{r} \cos\psi_g. \tag{4}$$

For measurement age $\Delta = t - t_g$, constant world velocity and yaw rate give the decision-time estimate.

$$\hat{E} = \widehat{E}_g + \Delta \widehat{V}_E, \quad \hat{N} = \widehat{N}_g + \Delta \widehat{V}_N. \tag{5}$$

$$\hat{\psi} = \operatorname{wrap}\left(\psi_g + \Delta \hat{r}\right). \tag{6}$$

$$\hat{u} = \widehat{V}_E \cos\hat{\psi} + \widehat{V}_N \sin\hat{\psi}. \tag{7}$$

$$\hat{v} = -\widehat{V}_E \sin\hat{\psi} + \widehat{V}_N \cos\hat{\psi}. \tag{8}$$

- Between valid GNSS records, the estimate propagates position using held body velocities and propagates heading with IMU yaw rate.

  - The estimator reads only delivered records and timestamps supplied by the policy interface.
    - It does not read simulation truth, environmental forces, actual thrust, motor mismatch, or future packets. It does not subtract the configured gyro bias.
- Valid GNSS acquisition age must stay within 0.5 s and valid IMU acquisition age within 0.1 s.
  - Until both records exist, or after either becomes stale, the policy commands neutral PWM and clears its integrators.
    - This startup behavior explains the small initial speed dip in sensor-based runs.
- This observer is timestamp extrapolation rather than an extended Kalman filter.
  - It does not model acceleration during GNSS latency or estimate sensor bias.
    - The oracle case removes this shared estimation limitation and separates it from the controller law.

1.2 Common motor proxy and limits

State feedback uses estimates of total thrust and yaw moment as motor states. A common internal motor proxy follows the requested PWM after a nominal 35 ms delay and a 0.2 s first-order lag. The proxy uses the nominal signed M200 map and does not receive acknowledgments for missed commands.

$$\dot{\hat{F}}_i = \frac{F_{i,\mathrm{nom}} - \hat{F}_i}{\tau}, \quad \tau = 0.2 \ \ \mathrm{s}. \tag{9}$$

- Requested force is clipped between approximately -20.068 N and 40.207 N per thruster.
  - The default controller converts bounded force to PWM using the inverse installed M200 lookup.
    - This feedforward conversion is identical for all three controllers. Motor mismatch, lag, slew limits, dropout, and timing remain in the plant.
- Conditional anti-windup freezes all stored integrators when either force request clips.
  - Speed, cross-track, and PID heading integrals are also bounded by 2 m, 12 m s, and 2 rad s, respectively.
    - The current integration candidate contributes to the command, but is stored only when feedback is ready and neither thruster clips.
- `--linear-pwm` replaces the inverse lookup with the nominal trim tangent.
  - This is an independent compensation experiment, not a change in feedback gains.
    - Keep the simulator nonlinear-thrust flag enabled when evaluating compensation on the full boat model.

2 Linear design model

The full nonlinear plant supplies the local design model. Linearization occurs at $u = U$, $v = r = \psi = 0$, zero environmental forcing, and $F_p = F_s = 7$ N. Wind remains an external disturbance because the feedback gains should not require its true value. The underlying marine equations and coupling follow the simulator's Fossen-style model [1].

- Added inertia is included in the design.
  - Nominal loaded mass is 21.9 kg; effective surge and sway masses are 24.09 kg and 32.85 kg, and effective yaw inertia is approximately 5.048 kg m².
    - The longitudinal drag derivative is $d_u = d_1 + 2d_2U = 24$ N s/m. Lateral and yaw damping are $d_v = 35$ N s/m and $d_r = 12$ N m s.
- The equilibrium total force is $S_0 = d_1U + d_2U\,|U| = 14$ N.
  - Motor sum and moment dynamics have the same time constant τ.
    - Saturation, deadband, variable map slope, waves, noise, and timing errors are evaluated in simulation rather than inserted into the linear design matrices.

2.1 Surge subsystem

Define the augmented surge state $z_s = (\delta u, \delta S, I_u)^T$, where $\delta S = F_p + F_s - S_0$ and $\dot{I}_u = \delta u$. Its control input is $\delta S_c = S_c - S_0$.

$$\dot{z_s} = A_s z_s + B_s \delta S_c. \tag{10}$$

$$A_s = \begin{pmatrix} -\frac{d_u}{m_x} & \frac{1}{m_x} & 0 \\ 0 & -\frac{1}{\tau} & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad B_s = \begin{pmatrix} 0 \\ \frac{1}{\tau} \\ 0 \end{pmatrix}. \tag{11}$$

2.2 Sway-yaw subsystem

The lateral augmented state is $z_l = \left(e_y, \psi, v, r, M, I_y\right)^T$, where $M = \frac{b(F_s - F_p)}{2}$ and $\dot{I}_y = e_y$. The input is desired yaw moment M_c.

$$\dot{z_l} = A_l z_l + B_l M_c. \tag{12}$$

$$A_l = \begin{pmatrix} 0 & U & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{d_v}{m_y} & -m_x\frac{U}{m_y} & 0 & 0 \\ 0 & 0 & (m_x - m_y)\frac{U}{I_z} & -\frac{d_r}{I_z} & \frac{1}{I_z} & 0 \\ 0 & 0 & 0 & 0 & -\frac{1}{\tau} & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}. \tag{13}$$

$$B_l = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \frac{1}{\tau} \\ 0 \end{pmatrix}. \tag{14}$$

- The surge and lateral controllability ranks are 3 and 6, matching their state counts.
  - The implemented matrix coefficients are checked against central finite differences of the full nonlinear hull.
    - Separating the local subsystems does not remove nonlinear surge/sway/yaw coupling from the simulator.

3 Three controller designs

The common model and interfaces allow three different feedback choices. PID organizes feedback into familiar loops. LQR chooses gains from a quadratic cost, while pole placement chooses the local response rates explicitly. Both state-feedback controllers use the same states and integral definitions, so their different results arise from their gain choices.

3.1 Cascaded PID

The proportional-integral-derivative (PID) controller regulates surge speed and heading. An outer cross-track proportional-integral loop converts line error into desired heading, and the measured sideslip estimate compensates the difference between hull heading and direction of travel.

$$\beta = \operatorname{atan2}\,(\hat{v}, \max(\hat{u}, 0.2)). \tag{15}$$

$$\psi_d = \operatorname{clip}\,\left(-k_y e_y - k_{Iy} I_y - \beta, -35°, 35°\right). \tag{16}$$

$$e_\psi = \operatorname{wrap}\,\left(\hat{\psi} - \psi_d\right). \tag{17}$$

$$S_c = S_0 - k_{Pu}\delta u - k_{Iu} I_u - k_{Du} q. \tag{18}$$

$$M_c = -k_{P\psi} e_\psi - k_{I\psi} I_\psi - k_{D\psi}\hat{r}. \tag{19}$$

- The speed derivative is filtered with time constant 0.1 s.
  - With tick interval h, the filter coefficient is $\alpha = \frac{h}{0.1+h}$.
    - Update $q_k = q_{k-1} + \alpha\left(\frac{\delta u_k - \delta u_{k-1}}{h} - q_{k-1}\right)$. Heading derivative action uses measured yaw rate and therefore avoids a derivative kick from a heading-reference step.
- The cross-track loop is slower than the heading loop.
  - Its heading clamp prevents large line errors from requesting a near-sideways course.
    - The design requires forward motion. The denominator floor regularizes the sideslip computation; it does not establish zero-speed controllability.
- PID tuning used an initial engineering choice and a heading-damping adjustment after an exploratory full-model run.
  - The final gains below were frozen for all cases and seeds.
    - No claim is made that these gains are globally optimal or that all three methods had equal optimization effort.

Loop	Gains	Interpretation
Surge PID	35, 12, 4	Force per speed, integrated speed, and speed derivative.
Heading PID	12, 1, 15	Moment per heading, integrated heading, and yaw rate.

Loop	Gains	Interpretation
Cross-track PI	0.18, 0.005	Heading per cross-track error and integrated cross-track error.

3.2 LQR with integral action

The linear quadratic regulator (LQR) minimizes an infinite-horizon quadratic cost for each augmented subsystem. The state integral therefore enters the cost along with the physical states. SciPy solves the continuous algebraic Riccati equation [2].

$$J = \int_0^\infty \left(z^T Q z + a^T R a\right) \mathrm{d}t. \tag{20}$$

$$A^T P + P A - P B R^{-1} B^T P + Q = 0. \tag{21}$$

$$K = R^{-1} B^T P, \quad a = -K z. \tag{22}$$

- The surge weights are $Q_s = \mathrm{diag}(25, 0.0025, 4)$ and $R_s = 0.01$.
  - The lateral weights are $Q_l = \mathrm{diag}(4, 25, 11.111111, 25, 0.04, 0.111111)$ and $R_l = 0.0625$.
    - These engineering weights normalize speed, line error, angular motion, motor effort, and integral accumulation; they do not come from a measured optimum.
- Commands satisfy $S_c = S_0 - K_s z_s$ and $M_c = -K_l z_l$.
  - The integral coordinates create constant force and heading biases needed to reject steady disturbances.
    - The resulting stability statement applies locally to the unconstrained continuous linear model. The implemented sampled nonlinear system is checked by simulation.

3.3 Pole placement with integral action

Pole placement uses the same augmented models but requests eigenvalues directly. SciPy calculates K so that the eigenvalues of $A - BK$ match the specified poles [3].

- Surge poles are $(-0.5, -1.3, -5.5)$ s⁻¹.
  - Lateral poles are $(-0.25, -0.4, -0.7, -1.2, -3.0, -5.5)$ s⁻¹.
    - The slower lateral modes allow line and integral recovery; faster modes govern yaw and motor response. Distinct poles avoid repeated-pole restrictions on each single-input subsystem.
- Commands use the same $S_c = S_0 - K_s z_s$ and $M_c = -K_l z_l$ form.
  - The test checks the achieved eigenvalues numerically.
    - Pole placement controls local response rates but does not itself minimize effort or guarantee robustness to latency, saturation, or model error.

3.4 Numerical gains and local poles

The following gains are loaded directly from the saved design. Their units follow the state coordinates and force or moment output; gains across different coordinates should not be compared as dimensionless numbers.

3.4.1 Surge gains

Method	Gain row in the declared state order
LQR + integral	36.083, 0.36, 20
Pole placement	20.572, 0.261, 17.224

LQR closed-loop poles in s⁻¹. −5.211, −2.226, −0.358.
Pole placement closed-loop poles in s⁻¹. −5.5, −1.3, −0.5.

3.4.2 Lateral gains

Method	Gain row in the declared state order
LQR + integral	13.469, 44.028, −1.054, 18.748, 0.768, 1.333
Pole placement	11.855, 37.045, −1.752, 13.023, 0.521, 1.313

LQR closed-loop poles in s⁻¹. −5.056 + 1.036i, −5.056 - 1.036i, −0.186, −0.524, −0.73 + 0.085i, −0.73 - 0.085i.
Pole placement closed-loop poles in s⁻¹. −5.5, −3, −1.2, −0.25, −0.4, −0.7.

4 Comparison protocol

The controller designs now meet the same simulator boundary, so the experiment varies plant assumptions while retaining the gains and reference. Full-model wind sweeps measure disturbance rejection. The cumulative flag ladder shows what happens as model features are added, and ablations show what happens when a feature group is removed from the full plant.

- Every run lasts 40 s and uses a 5 ms maximum integration step, 20 ms logging, and a 20 ms controller tick.
  - The initial condition is $(E, N, \psi, u, v, r) = (0, 2, 0.12, 0.8, 0, 0)$ in SI units.
    - Seeds 792, 793, and 794 supply paired gust, measurement, dropout, missed-command, and jitter streams. The policies produce their own commands; commands are not forced to match across controllers.
- All gains, limits, estimator constants, and nominal motor-proxy parameters remain fixed.
  - The primary study uses installed M200 inverse-map compensation for every case.
    - In the linear-thrust ladder stages and ablation, the same physical inverse map feeds a tangent actuator approximation. That deliberate model mismatch is part of the comparison, not a retuning of the controller.
- The full plant is preset `07_uncertainty`.
  - Its 5 m/s crosswind points north. Gust standard deviation is 0.5 m/s; sway wave force is 1 N and yaw wave moment is 0.1 N m with a 4 s period.
    - At zero heading, steady lateral projected-area wind force is approximately 4.59 N at 5 m/s, 11.76 N at 8 m/s, and 18.38 N at 10 m/s. Actual apparent-wind forcing changes with vessel motion and heading.

4.1 Cases and simulator flags

The following table records the changed fields. Every omitted field keeps its full-model value. The heavy-drag case adds parameter uncertainty and nonzero quadratic lateral/yaw damping rather than removing flags.

Case	Difference from the full model
`full_calm`	Disable wind, gusts, apparent wind, wind yaw moment, and waves.
`full_wind5`	Full model at the nominal 5 m/s wind.
`full_wind8`, `full_wind10`	Change only mean wind speed to 8 or 10 m/s.
`linear_hull`	Disable nonlinear kinematics, coupling, and drag. Keep added inertia.
`linear_thrust`	Disable nonlinear thrust, deadband, and reverse asymmetry. Keep saturation and fixed inverse-map compensation.
`ideal_motor`	Disable motor lag, force slew limit, and gain mismatch.
`ideal_timing`	Disable command delay, jitter, missed commands, and watchdog. Retain 50 Hz decisions.
`oracle_feedback`	Enable perfect-state feedback. The motor proxy and gains stay fixed.
`no_added_mass`	Disable added inertia.
`heavy_drag`	Mass ×1.15, additional inertia scale ×1.1, all four drag scales ×1.2; add sway quadratic damping 10 N s²/m² and yaw quadratic damping 3 N m s².

- The separate ladder begins with the original `00_linear` through `07_uncertainty` presets.
  - Command sampling is forced on in all eight stages because a feedback policy requires explicit decision ticks.
    - All later flags keep their original cumulative meaning. No controller is redesigned for a ladder stage.
- Ablation differences are conditional on the full model.
  - They isolate the selected group under the remaining enabled effects, rather than proving an additive contribution.
    - For example, removing motor lag also changes how sensor noise reaches the hull. Smaller error in one ablation does not establish a more accurate physical model.

4.2 Metrics

The main score is cross-track root-mean-square (RMS) error computed from internal hull truth. Sensor errors affect feedback but do not define the ground-truth score. Speed error is measured against surge speed, which is the declared controller reference rather than ground speed along east.

$$\mathrm{RMS}_y = \sqrt{\frac{1}{n}\sum_{k=1}^{n}(N_k - N_d)^2}. \tag{23}$$

$$\mathrm{RMS}_u = \sqrt{\frac{1}{n}\sum_{k=1}^{n}(u_k - U)^2}. \tag{24}$$

$$\mathrm{Effort} = \frac{1}{T}\int_0^T \left(F_p^2 + F_s^2\right)\mathrm{d}t. \tag{25}$$

- Whole-run RMS includes the initial 2 m recovery.
  - Final-ten-second RMS isolates sustained tracking after the transient.
    - The score uses all 20 ms truth samples; report plots retain every fifth sample for readability.
- Settling time is the first logged time after which $|e_y| \le 0.25$ m and $|\delta u| \le 0.1$ m/s remain true through the end.
  - A run can have low RMS but no settling time if gusts cause a late threshold crossing.
    - Aggregated settling values use only settled runs. The settled count is shown explicitly.
- Allocation-limited fraction counts controller ticks whose raw candidate forces exceed the nominal signed bounds, including startup ticks whose final command is overridden to neutral.
  - Mean squared thrust uses actual simulated motor force.
    - It is an effort proxy, not electrical energy. Voltage, current, battery state, and propeller inflow are not simulated.

5 Full-model results

The full-model tests show how the controllers behave when the boat retains its nonlinearities and the navigation feedback includes realistic delivery imperfections. The figures use seed 792; the tables include all three seeds so that one favorable gust sequence cannot determine the comparison.

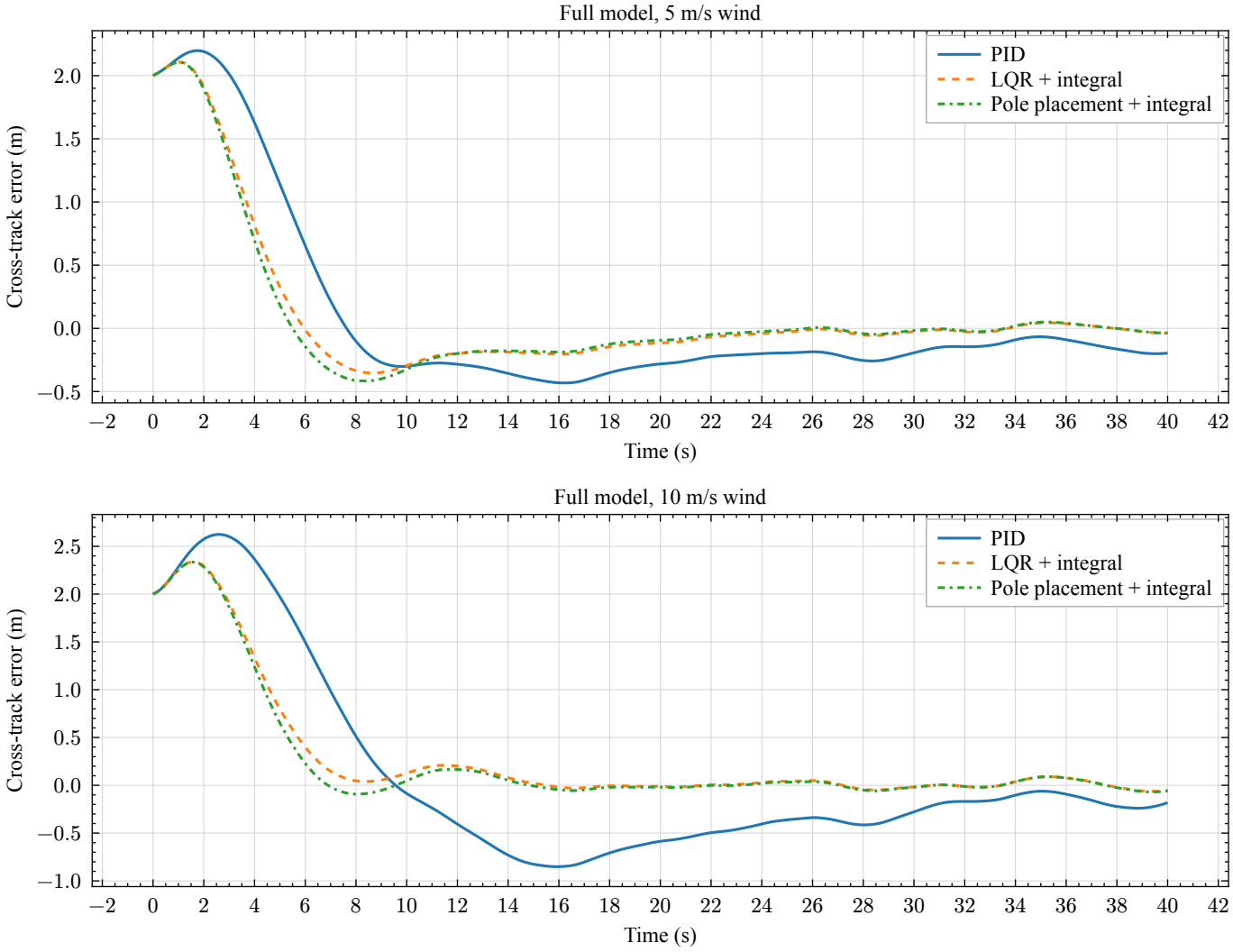


Figure 2: Cross-track recovery in the full model, seed 792.

Wind	Controller	Whole-run RMS (m)	Tail RMS (m)	Tail speed RMS (m/s)	Effort (N^2)
Calm	PID	0.683	0.163	0.006	102.7
Calm	LQR + integral	0.583	0.011	0.005	110.5
Calm	Pole placement + integral	0.575	0.004	0.004	112.9
5 m/s	PID	0.75	0.181	0.007	113
5 m/s	LQR + integral	0.592	0.034	0.006	120.1
5 m/s	Pole placement + integral	0.583	0.034	0.005	122.6
8 m/s	PID	0.884	0.21	0.009	136.8
8 m/s	LQR + integral	0.636	0.049	0.008	144.1
8 m/s	Pole placement + integral	0.624	0.048	0.008	146.7
10 m/s	PID	1.018	0.229	0.013	171.5
10 m/s	LQR + integral	0.696	0.056	0.01	179.7
10 m/s	Pole placement + integral	0.682	0.053	0.011	182.6

At 5 m/s wind, the mean tail line errors are 0.181 m for PID, 0.034 m for LQR, and 0.034 m for pole placement. At 10 m/s wind, LQR reduces PID's tail error by 75.6% while increasing mean squared thrust by 4.8%.

The fixed PID has greater residual line error than either state-feedback design in this envelope. LQR and pole placement produce very similar tail errors, so their small numerical differences should be judged together with transient recovery and thrust effort.

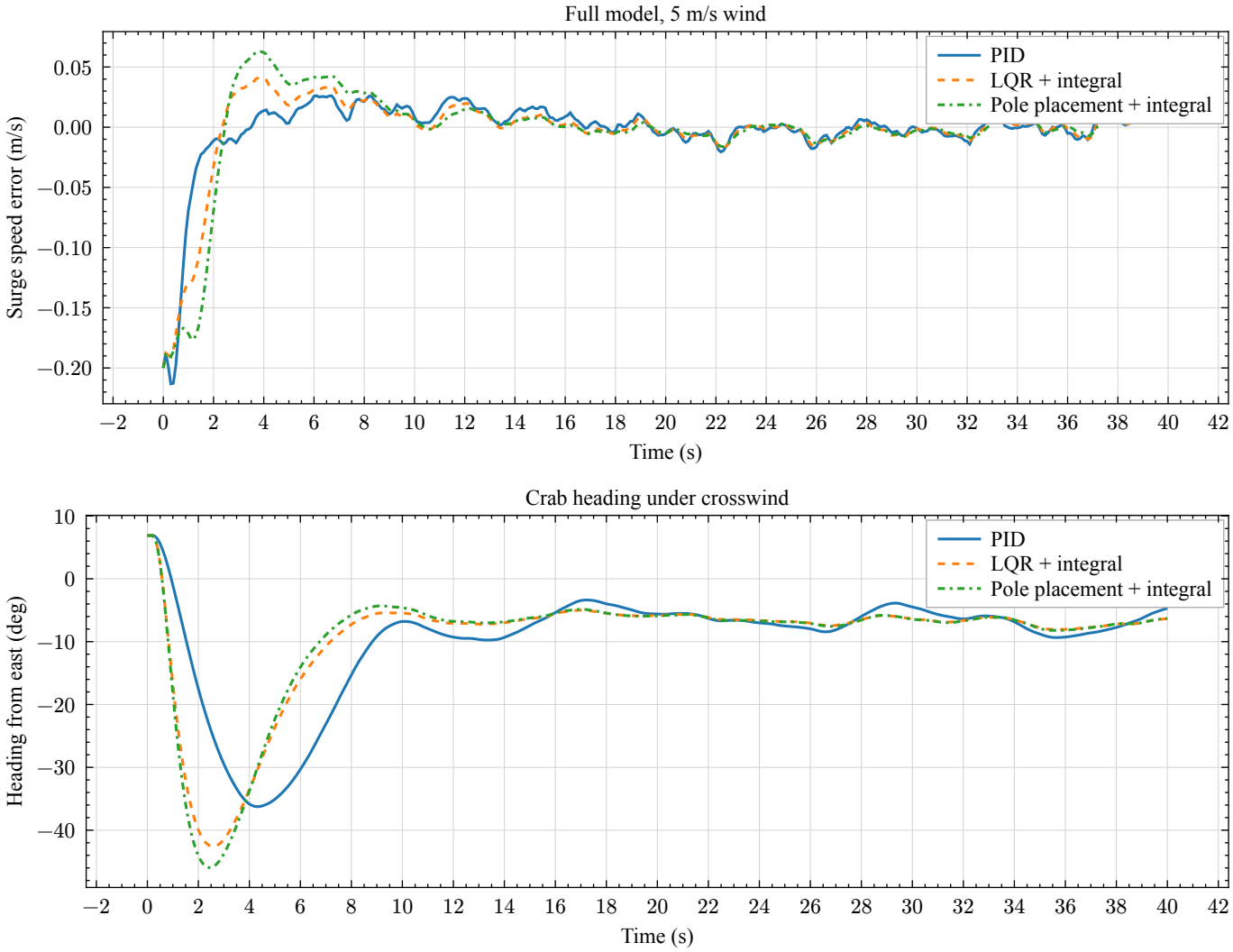


Figure 3: Speed regulation and crab heading, seed 792.

- The vessel points into the crosswind while holding the east-going line.
  - This heading is expected because the thrusters cannot push directly sideways.
    - Integral action and the coupling-aware feedback create the required yaw bias. PID additionally applies explicit sideslip compensation in its guidance loop.
- The small speed transient includes sensor startup and actuator delay.
  - Subsequent speed errors reflect gusts, wave forcing, allocation, and the controller's surge gains.
    - The local linearization includes surge drag slope and motor lag but not nonlinear sway/yaw energy exchange away from trim.

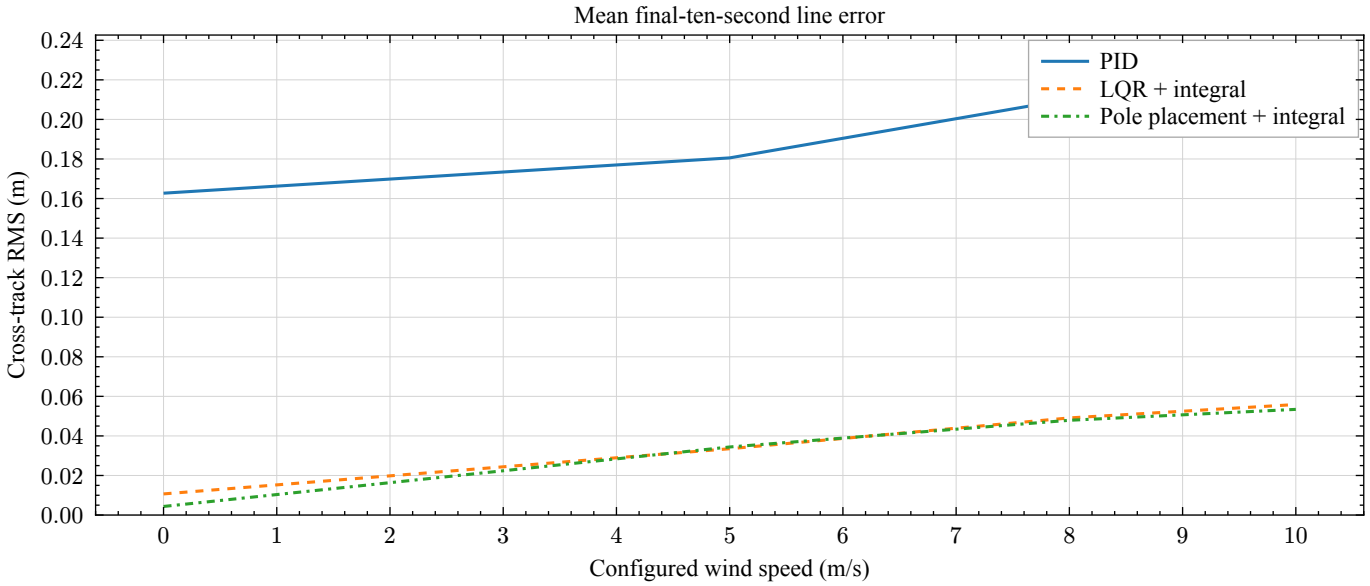


Figure 4: Mean tail line error across three paired seeds.

The calm point disables waves and gusts along with wind. The 5, 8, and 10 m/s points retain gusts and waves, so the jump from calm to 5 m/s includes all environmental forcing. The 5-to-10 m/s comparison changes only mean wind speed.

Wind	Controller	Tail RMS range (m)	Settled runs	Mean settling (s)
5 m/s	PID	0.142-0.209	3 / 3	30.6
5 m/s	LQR + integral	0.024-0.05	3 / 3	12.5
5 m/s	Pole placement + integral	0.026-0.051	3 / 3	12.4
8 m/s	PID	0.159-0.251	3 / 3	34.5
8 m/s	LQR + integral	0.037-0.071	3 / 3	6
8 m/s	Pole placement + integral	0.035-0.069	3 / 3	6.9
10 m/s	PID	0.173-0.276	3 / 3	34.7
10 m/s	LQR + integral	0.041-0.08	3 / 3	6.5
10 m/s	Pole placement + integral	0.037-0.075	3 / 3	7

5.1 Effort and actuator demand

The controller can improve line error by using more differential thrust. The effort and allocation records therefore accompany the tracking results. These records also identify whether an apparent response advantage depends on sustained clipping.

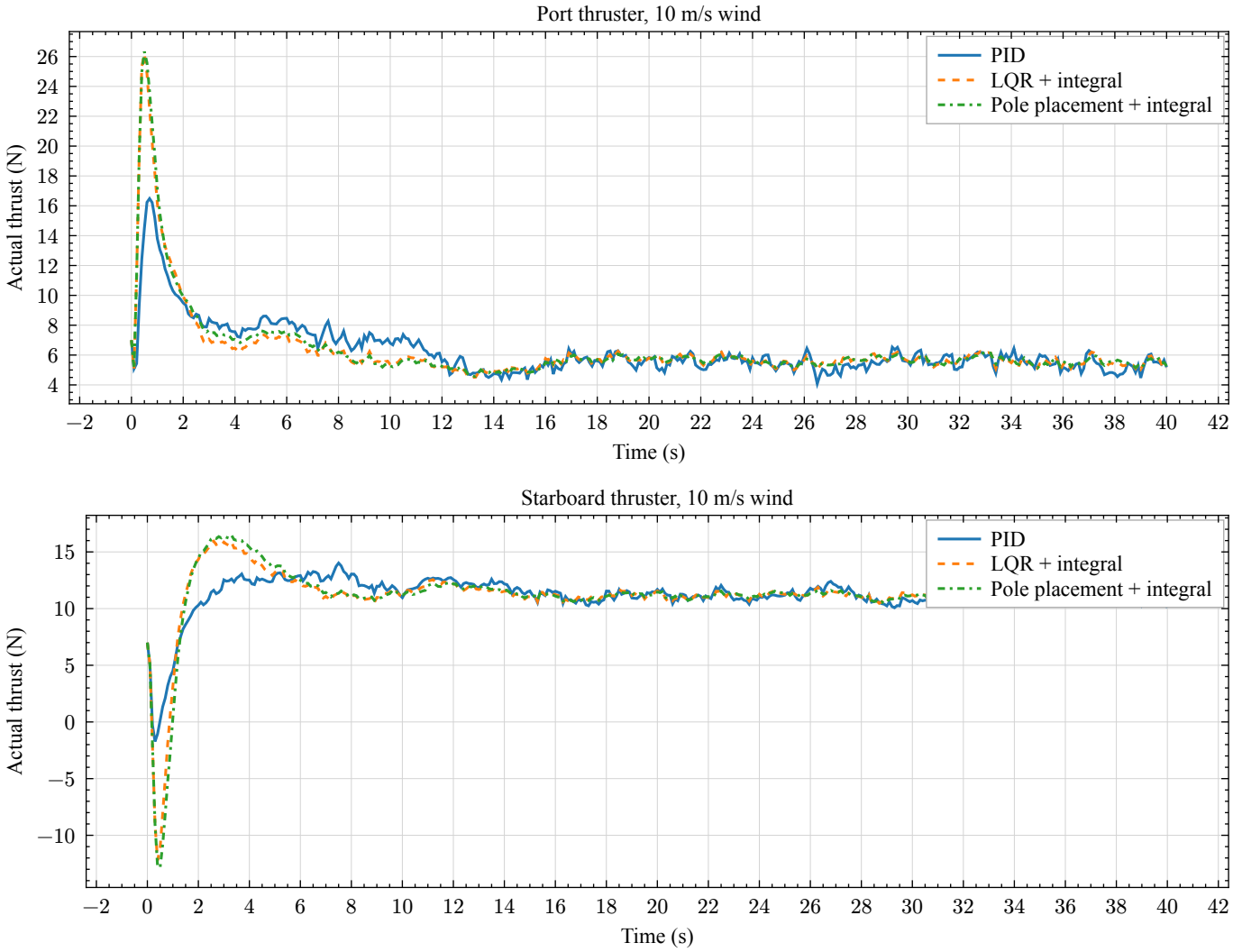


Figure 5: Actual M200 forces at 10 m/s wind, seed 792.

Controller at 10 m/s	Mean allocation-limited ticks (%)	Maximum actual force (N)
PID	0	16.6
LQR + integral	0.5	26.31
Pole placement + integral	0.5	26.36

6 Progressive realism and ablations

The preceding wind results establish performance on the full model. The flag comparisons now explain which simplifications affect that performance, while preserving the same controller gains and M200 compensation.

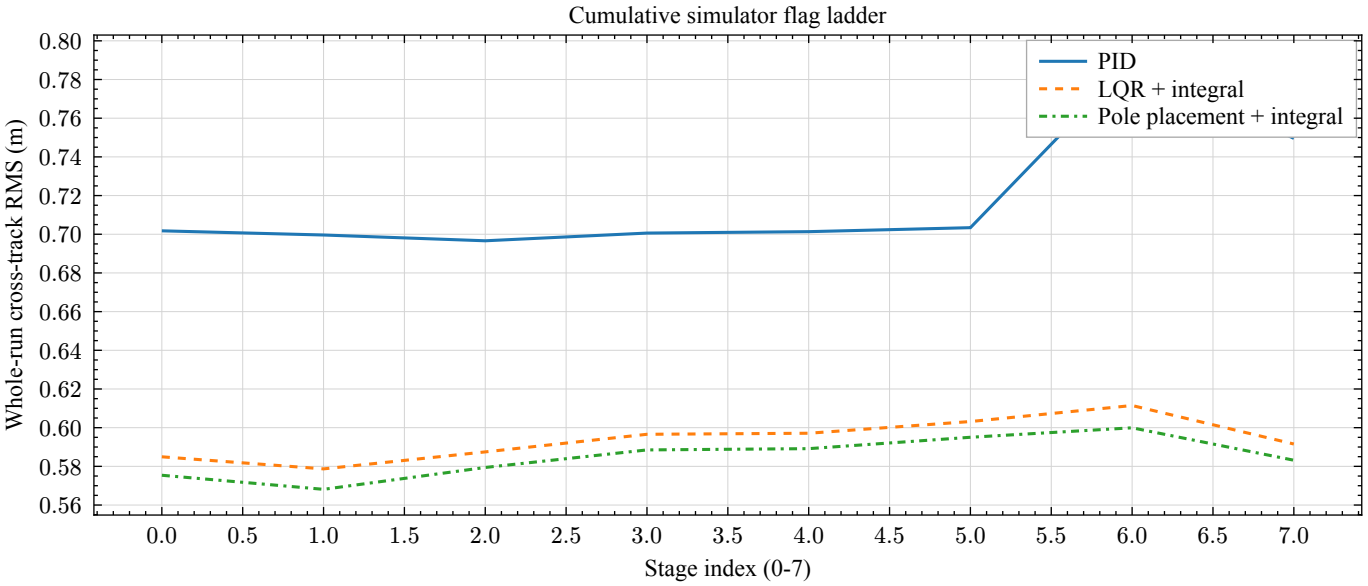


Figure 6: Whole-run line error as cumulative features are added.

Index	Preset	New feature group
0	`00_linear`	Local hull and tangent actuator; perfect hull state.
1	`01_hull`	Nonlinear hull rotation, coupling, and surge drag.
2	`02_m200`	Bounded signed M200 lookup, neutral deadband, reverse asymmetry.
3	`03_motor`	Motor lag and force slew limit.
4	`04_timing`	35 ms command delay.
5	`05_sensors`	Noise, bias, delay, antenna offset, camera occlusion; remove perfect-state feedback.
6	`06_disturbances`	Wind, gusts, apparent wind, wind moment, prescribed waves.
7	`07_uncertainty`	Added inertia, motor mismatch, jitter, missed commands, watchdog, sensor dropout.

The ladder does not imply that increasing fidelity must increase error. A controller can accidentally perform better on a more detailed plant because of changed damping, gain, or phase. The following full-model ablations make that dependence explicit.

Case	PID tail RMS (m)	LQR tail RMS (m)	Pole-placement tail RMS (m)
`full_wind5`	0.181	0.034	0.034
`linear_hull`	0.179	0.034	0.035
`linear_thrust`	0.177	0.034	0.035
`ideal_motor`	0.165	0.034	0.034
`ideal_timing`	0.18	0.034	0.034
`oracle_feedback`	0.177	0.034	0.035
`no_added_mass`	0.168	0.05	0.05
`heavy_drag`	0.177	0.026	0.028

- Small hull and thrust-ablation differences indicate that this line-tracking maneuver stays fairly close to the design envelope.
  - The inverse M200 lookup already compensates the static propulsion map in the full plant.
    - A reversal, aggressive speed transition, tight turn, or large saturation event can expose much larger nonlinear differences and should be a subsequent maneuver test.
- The oracle feedback case shows the incremental effect of the shared estimator under the configured sensor errors.
  - A small oracle advantage means wind rejection or gain structure is dominating this particular score.
    - It does not establish that sensor bias, RTK degradation, or longer outages are harmless. Those faults need dedicated sweeps.
- Added inertia and drag change both disturbance acceleration and the response to feedback.

- Their effect need not be monotonic. Higher damping can suppress a gust response while increasing required thrust.
  - The heavier-drag test therefore reports effort as well as tracking error in the saved metrics.

7 Verification, reproduction, and next design decision

The simulation evidence supports a controller choice for this operating envelope. Mathematical checks support the local design, while causal and numerical checks support the implementation. Those checks have different purposes and should remain separate when the controller moves onto a physical boat.

- Mathematical verification checks both augmented controllability ranks, the Riccati residual, all local closed-loop eigenvalues, and agreement of the nonlinear hull Jacobian with the design matrices.
  - Controller tests also check heading wrapping, lever-arm and latency correction, invalid-packet retention, neutral output without navigation feedback, force limits, anti-windup, and calm recovery.
    - The complete project test suite contains 21 tests. Code formatting, linting, and static typing accompany those tests. Halving the maximum integration step from 5 ms to 2.5 ms in the full 5 m/s case changes the sampled hull states by less than 7×10^{-9} in SI units for all three controllers.
- Comparison evidence uses paired seeds and frozen tuning.
  - Mean, minimum, maximum, and individual-seed metrics are retained, rather than only a winning example.
    - Three seeds describe this small sweep; they do not establish a broad probability of mission success.
- LQR with integral action is the candidate to take forward.
  - It gives comparable rejection to pole placement while making the effort-versus-error tradeoff explicit through Q and R.
    - Keep pole placement as a close benchmark and PID as the simpler fallback. This recommendation applies to the saved gains and moving-line task, not to every design in each controller family.

7.1 Run and inspect

Run the saved comparison from the project directory. The CLI accepts existing simulator overrides, so controller studies use the same configuration flags as open-loop simulation.

```
uv sync
uv run python -m blueboat_sim.controller_study
uv run python -m blueboat_sim.controller_study \
  --case full_wind5 --case full_wind10 --seeds 792,793,794 \
  --set flags.sensor_dropout=false \
  --data build/no-dropout/data --output build/no-dropout
uv run python -m blueboat_sim.controller_study \
  --case full_wind5 --linear-pwm \
  --data build/tangent-pwm/data --output build/tangent-pwm
typst compile controller_report.typ controller_report.pdf
```

- `controller_report/data/design.json` stores matrices, weights, gains, poles, and nominal parameters.
  - `summary.json` stores every resolved case, seed, metric, aggregate, and source hash; `traces.json` stores the representative plotting samples.
    - Per-run truth CSV, policy estimates, requested force, integral states, resolved configuration, and decision-delivery audits are saved under `build/controllers/`.
- Keep optional experiments in a separate `--data` directory.
  - The report compiles from the retained main-study JSON files.
    - Changing the default data directory replaces the report's input data and therefore requires a fresh report review.

7.2 Tests still needed before boat trials

The present controllers track a fixed east-going line at 1 m/s. They do not implement waypoint transitions, turning guidance, obstacle avoidance, station keeping, stereo-based navigation, or a full mission controller.

- Extend the reference tests to speed changes, reversals, curved paths, and large initial heading errors.
  - Test deliberate navigation outages, biased GNSS, longer command latency, and motor degradation.
    - Those cases should expose where fixed gains or the current minimal observer need replacement.
- Identify motor lag, installed thrust, mass/inertia, damping, and wind area from physical measurements.
  - Benchmark actual sensor and AGX Orin delivery timing.
    - The hardware controller should use measured uncertainty bounds and receive actuator acknowledgments where available.

References

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